

C Programming

Trainer : Rajiv K

Email: rajiv.kamune@sunbeaminfo.com



Agenda

- **Type Modifiers**
- **Signed / unsigned char**
- **Constants**
- **scanf**



Type Modifiers :

Type Modifiers : data type modifiers are the **keywords used to modify the original sign or length/range of values** that various primitive data types hold such as int, char, and double.

- **char**

- signed char (-128 to 127)
- unsigned char (0 to 255)

- **int/ long (32-bit)**

- signed int(-32,768 to 32,767 or -2,147,483,648 to 2,147,483,647)
- unsigned int(0 to 65,535 or 0 to 4,294,967,295)

- **short int**

- signed short (-32,768 to 32,767)
- unsigned short(0 to 65,535)

- **long long/ long (64-bit)**

- signed long (-9223372036854775808 to 9223372036854775807)
- unsigned long(0 to 18446744073709551615)

- float: $\pm 3.4E \pm 38$

- double: $\pm 1.7E \pm 308$



Type Modifiers Chart :

Type Modifier	Applicable Data Types
<u>signed</u>	int, char
<u>unsigned</u>	int, char
<u>short</u>	<u>int</u>
<u>long</u>	<u>int</u> , <u>double</u>
long long	int



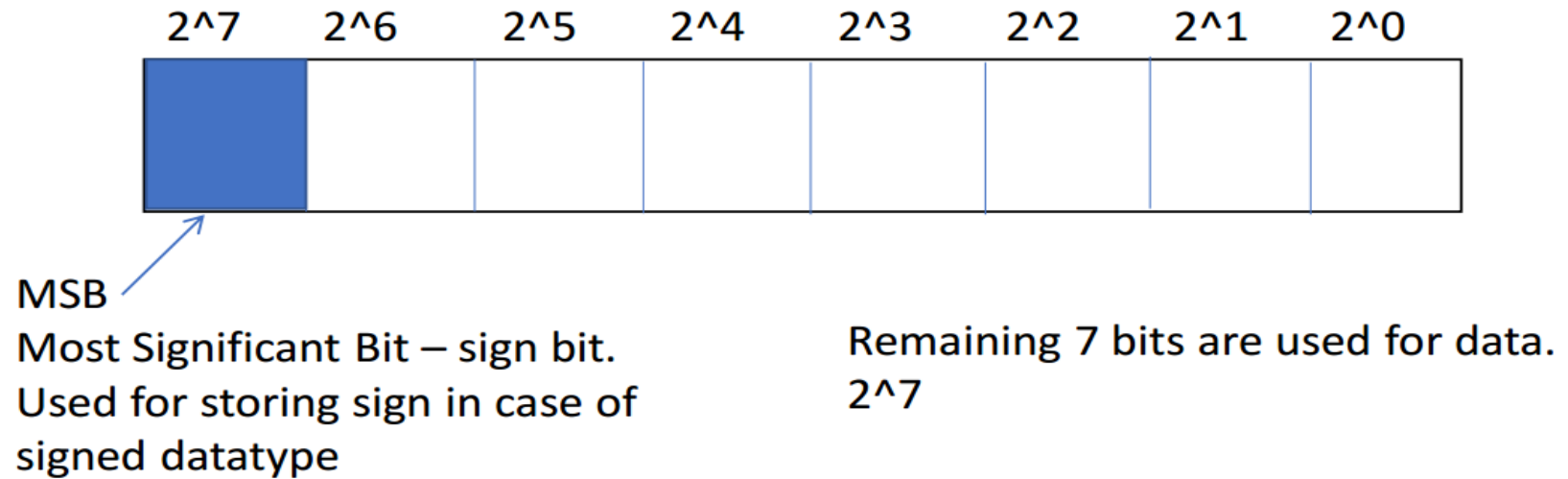
Type Modifiers :

C Basic Data Types	32-bit CPU		64-bit CPU	
	Size (bytes)	Range	Size (bytes)	Range
char	1	-128 to 127	1	-128 to 127
short	2	-32,768 to 32,767	2	-32,768 to 32,767
int	4	-2,147,483,648 to 2,147,483,647	4	-2,147,483,648 to 2,147,483,647
long	4	-2,147,483,648 to 2,147,483,647	8	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
long long	8	9,223,372,036,854,775,808 to 9,223,372,036,854,775,807	8	9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
float	4	3.4E +/- 38	4	3.4E +/- 38
double	8	1.7E +/- 308	8	1.7E +/- 308

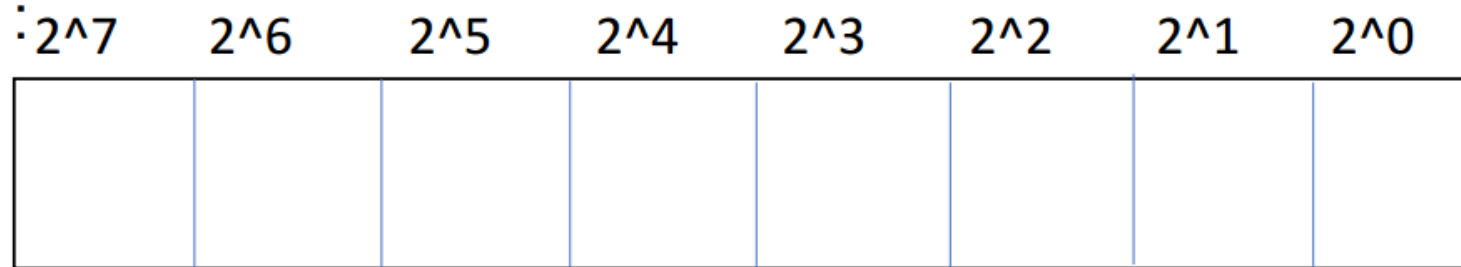


Signed / unsigned char :

- Signed char :



- Unsigned char :



MSB
Most Significant Bit – Not a sign bit

All 8 bits are used for data. 2^8



Constants

- **Numeric constants**
 - **Integer constants**
 - **Decimal**
 - **Octal**
 - **Hexadecimal**
 - **Real constants**
 - **floating point form**
 - **exponential form**
 - **Character constants**
 - **Single character constants**
 - **String constants**



Constant

- The constant is the entity whose value remains same during the execution of the program.

Literal constant:

- Integer literal constant:
 - If integer constant starts with 0 it is assumed to be in octal number system
- If integer constant starts with 0x or 0X then it is assumed to be in hexadecimal number system 0x23 or 0X23 is valid which means 23 in hexadecimal number system is equivalent to 19 in decimal.
- If integer constant has suffix character l, L, u, U, f and F, If integer constant is terminated with l, L then it is assumed to Long, if it is terminated with u or U then it is assumed to unsigned int. 23l is a long integer, 23u is a unsigned integer, f and F can be used with floating point not with integer
- Floating point literal constant
 - Must have at least one digit, it can be either positive or negative by default it is positive.
 - It should have decimal point
 - Floating point by default is considered to be double eg.23.45 is double.
 - It can be modified using length modifier f or F i.e 23.45 is written as 23.45f then it is considered of type float instead of double



Constant

- **Character literal constant.**
 - • Can have one or atmost two characters enclosed in single quotes eg.'A', 'b'
 - • Escape sequences(non printable character constant):-
- **String literal constants**
 - Sequence of the characters enclosed within the double quotes and terminated with '\0'
- **Qualified Constant:**
 - They are created using the const qualifier const char a='A';
- **Symbolic constant:**
 - They are created with help of #define preprocessor directive.



scanf

- **scanf is a standard library function used to receive formatted input from the standard input device**
- **(usually the screen)**
- **Requires header file `stdio.h`**
- **Used to input values from user.**
- **Format `scanf( formatstring [, value ] ... );`**
- **Should have one format specifier for each address passed, and types should match**
- **Do not use any char other than format specifiers in format string.**
- **To skip a char from input use `%*c`.**
- **Example - `scanf("%f",&temp);`**



Thank You

